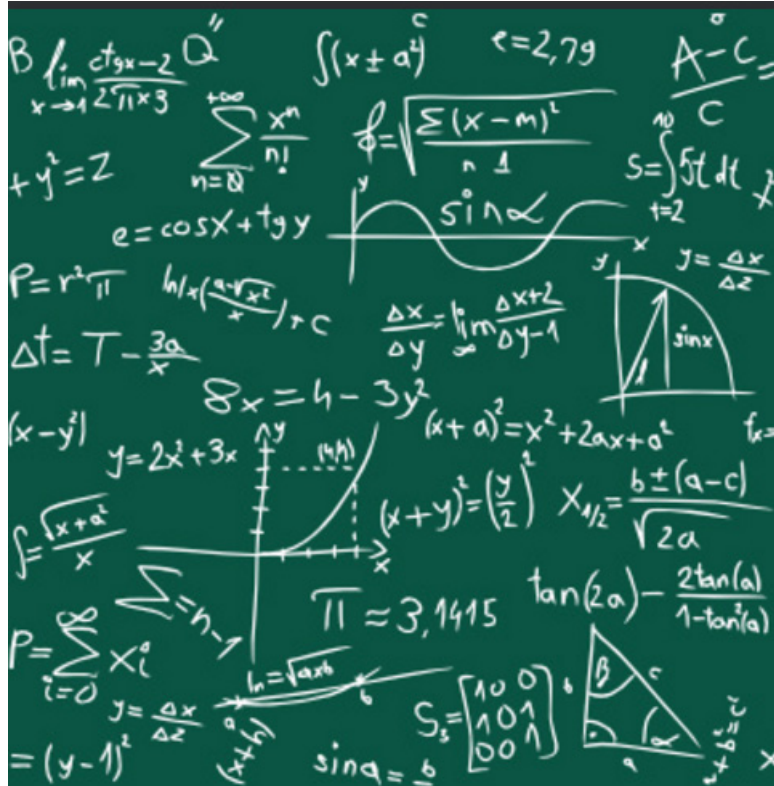


Cusps, inflection points and tangents



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

Exercise 1

Using the definition of derivative, determine the behavior at the origin of the following function:

$$f(x) = x^{\frac{4}{3}}$$

Let

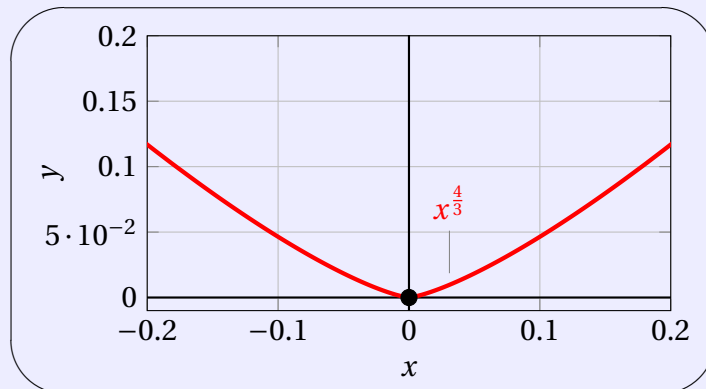
$$f(x) = x^{\frac{4}{3}}$$

then for $x_0 = 0$

$$\frac{f(h) - f(0)}{h} = \frac{h^{4/3}}{h} = h^{1/3}$$

and the limits are:

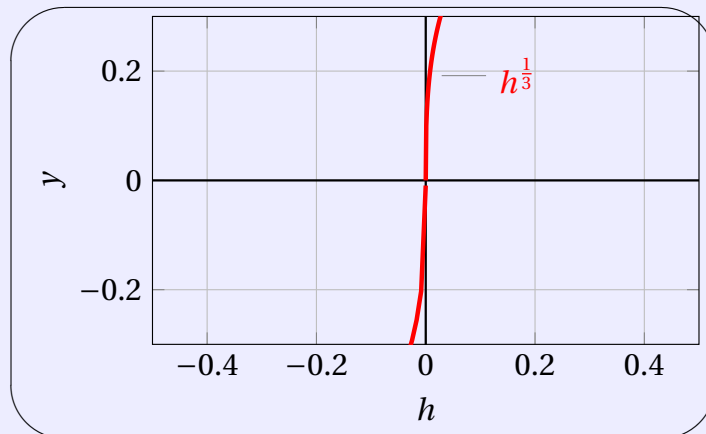
$$\lim_{h \rightarrow 0^-} h^{1/3} = 0 \text{ and } \lim_{h \rightarrow 0^+} h^{1/3} = 0 \text{ hence } \lim_{h \rightarrow 0} h^{1/3} = 0 \text{ and } f'(0) = 0.$$



The function has a **point with horizontal tangent** at $x_0 = 0$. The function:

$$f(h) = h^{\frac{1}{3}}$$

is a power with positive rational exponent $\frac{m}{n}$ with n odd (hence defined on all of $\mathbb{R}$) and m odd (hence an odd function). Its graph is:



Exercise 2

Using the definition of derivative, determine the behavior at the origin of the following function:

$$f(x) = x^{\frac{2}{3}}$$

Let

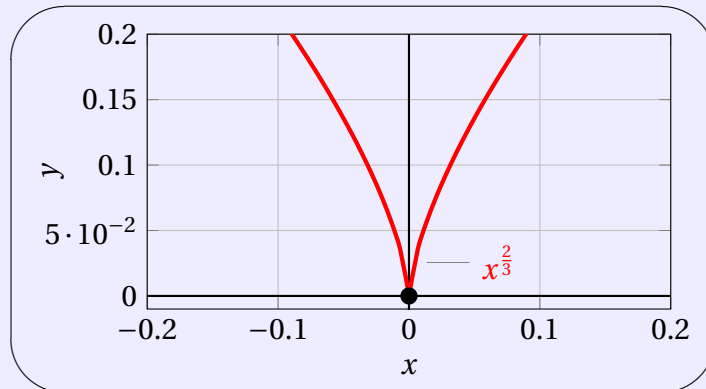
$$f(x) = x^{\frac{2}{3}}$$

then for $x_0 = 0$

$$\frac{f(h) - f(0)}{h} = \frac{h^{2/3}}{h} = \frac{1}{h^{1/3}}$$

and the limits are:

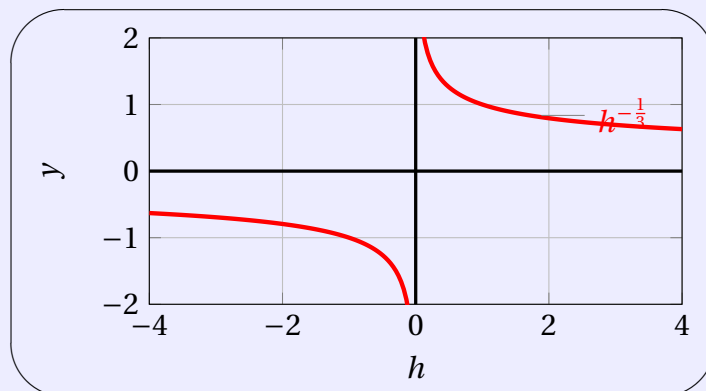
$$\lim_{h \rightarrow 0^-} \frac{1}{h^{1/3}} = -\infty \text{ and } \lim_{h \rightarrow 0^+} \frac{1}{h^{1/3}} = +\infty \text{ hence } f'_-(0) = -\infty \text{ and } f'_+(0) = +\infty$$



The function has a **cusp** at $x_0 = 0$. The function:

$$f(h) = h^{-\frac{1}{3}} = \frac{1}{h^{1/3}}$$

is a power with negative rational exponent $\frac{m}{n}$ with n odd (hence defined on all of $\mathbb{R}$) and m odd (hence an odd function). Its graph is:



Exercise 3

Using the definition of derivative, determine the behavior at the origin of the following function:

$$f(x) = x^{\frac{5}{2}}$$

Let

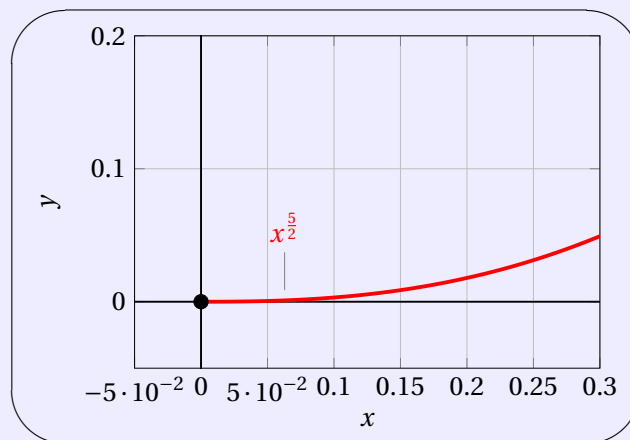
$$f(x) = x^{\frac{5}{2}}$$

then for $x_0 = 0$

$$\frac{f(h) - f(0)}{h} = \frac{h^{5/2}}{h} = h^{3/2}$$

and the limits are:

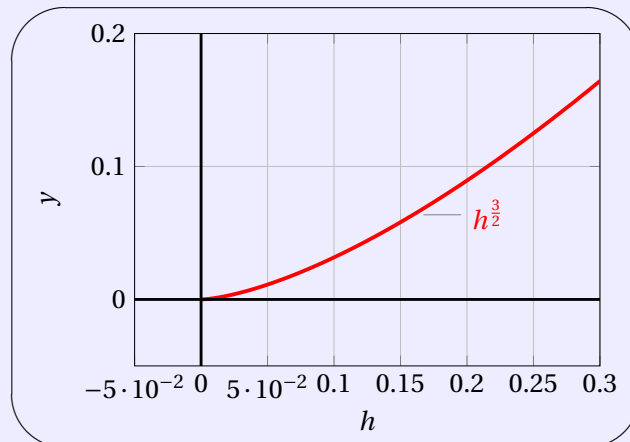
$$\lim_{h \rightarrow 0^+} h^{3/2} = 0.$$



The function has a **point with horizontal tangent** at $x_0 = 0$. The function:

$$f(h) = h^{\frac{3}{2}}$$

is a power with positive rational exponent $\frac{m}{n}$ with n even (hence defined only on $\mathbb{R}_+$). Its graph is:



Exercise 4

Using the definition of derivative, determine the behavior at the origin of the following function:

$$f(x) = x^{\frac{3}{2}}$$

Let

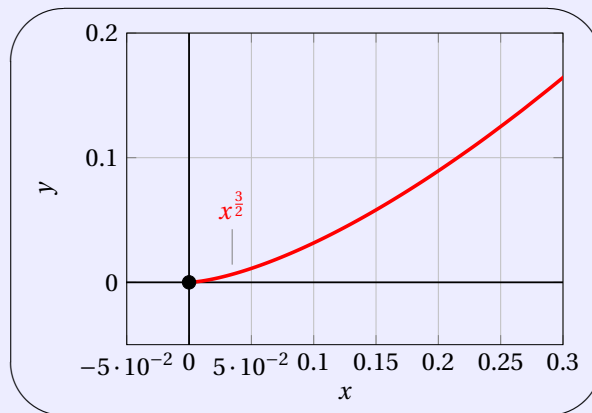
$$f(x) = x^{\frac{3}{2}}$$

then for $x_0 = 0$

$$\frac{f(h) - f(0)}{h} = \frac{h^{3/2}}{h} = h^{1/2}$$

and the limits are:

$$\lim_{h \rightarrow 0^+} h^{1/2} = 0.$$



The function has a **point with horizontal tangent** at $x_0 = 0$. The function:

$$f(h) = h^{\frac{1}{2}}$$

is a power with positive rational exponent $\frac{m}{n}$ with n even (hence defined only on $\mathbb{R}_+$). Its graph is:

